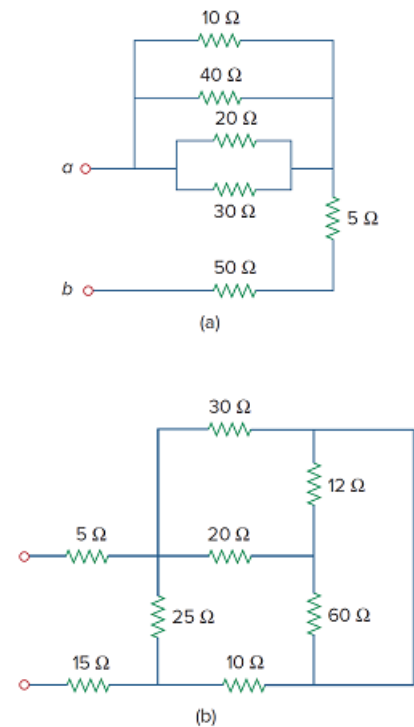


Problem

Find the equivalent resistance at terminals a - b of each circuit in Fig. 2.109.

Figure 2.109:



Step-by-step solution

Step 1 of 13

(a)

Refer to Figure 2.109 in the textbook.

In the circuit, resistors $20\ \Omega$, $30\ \Omega$ and $10\ \Omega$, $40\ \Omega$ are in connected in parallel.

Calculate the parallel equivalent resistance R_{eq1} for $20\ \Omega$, $30\ \Omega$.

$$\begin{aligned} R_{eq1} &= \frac{(20\ \Omega)(30\ \Omega)}{20\ \Omega + 30\ \Omega} \\ &= \frac{600}{50}\ \Omega \\ &= 12\ \Omega \end{aligned}$$

Calculate the parallel equivalent resistance R_{eq2} for $10\ \Omega$, $40\ \Omega$.

$$\begin{aligned} R_{eq2} &= \frac{(10\ \Omega)(40\ \Omega)}{10\ \Omega + 40\ \Omega} \\ &= \frac{400}{50}\ \Omega \\ &= 8\ \Omega \end{aligned}$$

Step 2 of 13

Draw the modified circuit diagram.

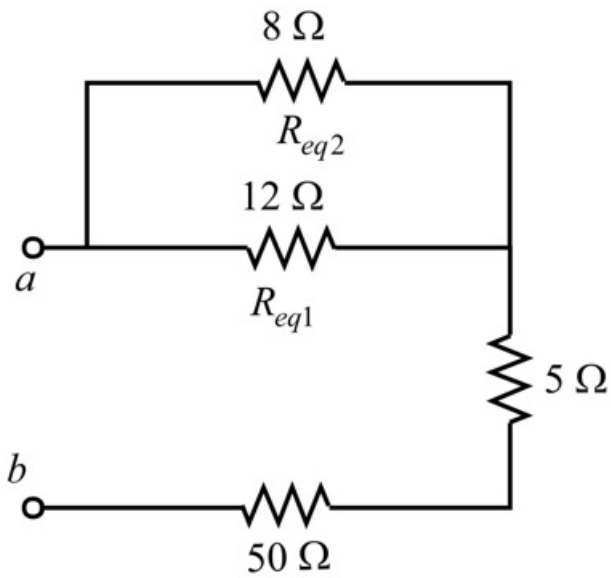


Figure 1

Step 3 of 13

In Figure 1, resistors $12\ \Omega$ and $8\ \Omega$ are connected in parallel. Determine its parallel equivalent resistance, R_{eq3} .

$$\begin{aligned} R_{eq3} &= \frac{(12\ \Omega)(8\ \Omega)}{12\ \Omega + 8\ \Omega} \\ &= \frac{96}{20}\ \Omega \\ &= 4.8\ \Omega \end{aligned}$$

Step 4 of 13

Redraw the modified circuit diagram.

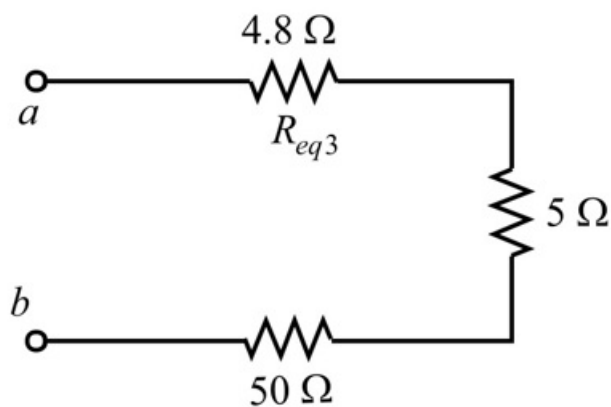


Figure 2

Step 5 of 13

In Figure 2, resistors $4.8\ \Omega$, $5\ \Omega$ and $50\ \Omega$ are connected in series. Determine its equivalent resistance, R_{ab} .

$$\begin{aligned} R_{ab} &= 4.8\ \Omega + 5\ \Omega + 50\ \Omega \\ &= 59.8\ \Omega \end{aligned}$$

Therefore, the equivalent resistance at the terminals a-b, R_{ab} is $\boxed{59.8\ \Omega}$.

Step 6 of 13

(b)

Refer to Figure 2.109 in the textbook.

The circuit can be re-drawn as follow:

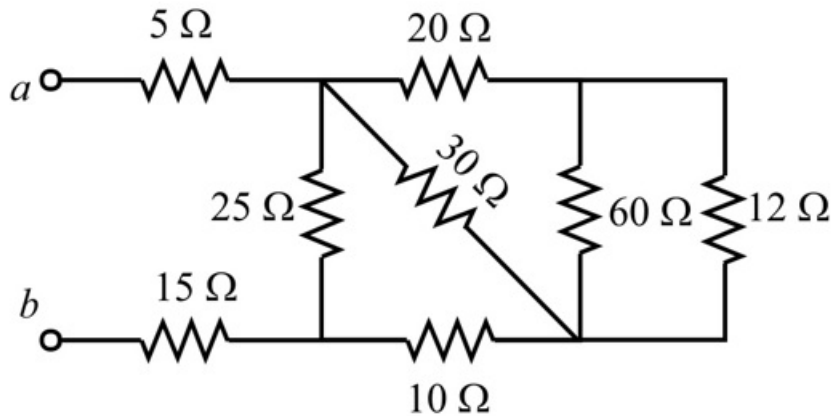


Figure 3

Step 7 of 13

In Figure 3, resistors $12\ \Omega$ and $60\ \Omega$ are in connected in parallel.

Calculate the parallel equivalent resistance R_{eq1} for $12\ \Omega$ and $60\ \Omega$.

$$\begin{aligned} R_{eq1} &= \frac{(12\ \Omega)(60\ \Omega)}{12\ \Omega + 60\ \Omega} \\ &= \frac{720\ \Omega}{72\ \Omega} \\ &= 10\ \Omega \end{aligned}$$

Draw the modified circuit diagram.

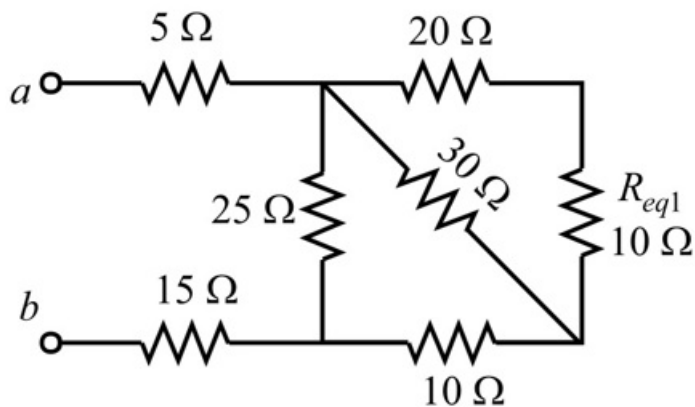


Figure 4

Step 8 of 13

In Figure 4, resistors $20\ \Omega$ and $10\ \Omega$ are in series. Determine its equivalent series resistance,

$$R_{eq2}$$

$$R_{eq2} = 20\ \Omega + 10\ \Omega$$

$$= 30\ \Omega$$

Draw the modified circuit diagram.

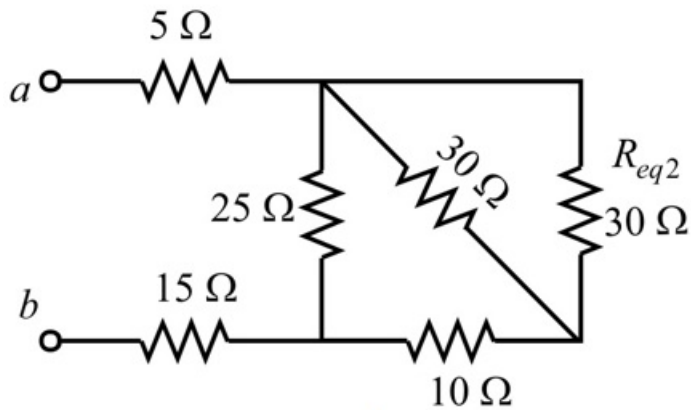


Figure 5

Step 9 of 13

In Figure 5, two $30\ \Omega$ resistors are connected in parallel. Calculate the parallel equivalent resistance, R_{eq3} .

$$R_{eq3} = \frac{(30\ \Omega)(30\ \Omega)}{30\ \Omega + 30\ \Omega}$$

$$= \frac{900\ \Omega}{60\ \Omega}$$

$$= 15\ \Omega$$

Draw the modified circuit diagram.

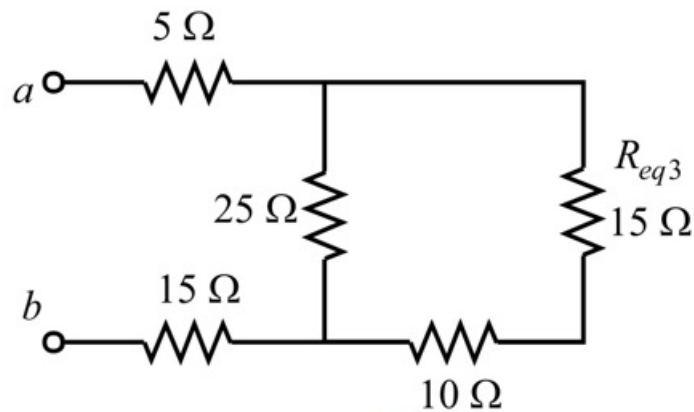


Figure 6

Step 10 of 13

In Figure 6, resistors $15\ \Omega$ and $10\ \Omega$ are in connected in series. Calculate the series equivalent resistance, R_{eq4} .

$$\begin{aligned} R_{eq4} &= 15\ \Omega + 10\ \Omega \\ &= 25\ \Omega \end{aligned}$$

Draw the modified circuit diagram.

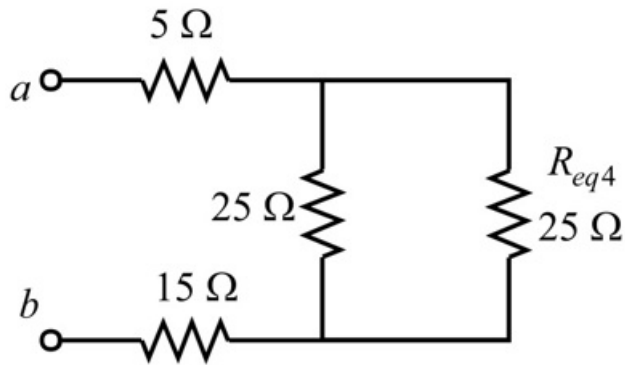


Figure 7

Step 11 of 13

In Figure 7, two $25\ \Omega$ resistors are connected in parallel. Calculate the parallel equivalent resistance, R_{eq5} .

$$\begin{aligned} R_{eq5} &= \frac{(25\ \Omega)(25\ \Omega)}{25\ \Omega + 25\ \Omega} \\ &= \frac{625\ \Omega}{50\ \Omega} \\ &= 12.5\ \Omega \end{aligned}$$

Step 12 of 13

Draw the modified circuit diagram.

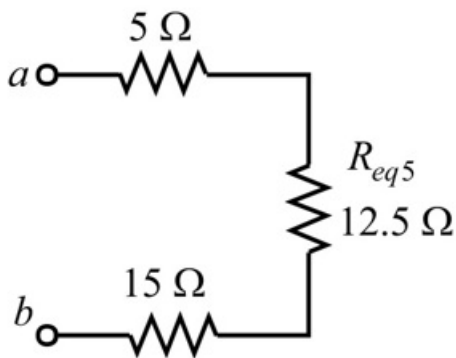


Figure 8

Step 13 of 13

In Figure 8, resistors $5\ \Omega$, $15\ \Omega$ and $12.5\ \Omega$ are connected in series. Calculate its series equivalent resistance R_{ab} .

$$\begin{aligned} R_{ab} &= 5\ \Omega + 15\ \Omega + 12.5\ \Omega \\ &= 32.5\ \Omega \end{aligned}$$

Therefore, the equivalent resistance at the terminals a-b, R_{ab} is $\boxed{32.5\ \Omega}$.